

REGRESSION ASSIGNMENT

Insurance Charges Prediction Using Machine Learning

Data Science / Machine Learning Project

Prepared By	Sanjay R
Problem Type	Supervised Learning – Regression
Target Variable	charges
Evaluation Metric	R ² Score

Final Model: Random Forest Regressor

Best Test R² Score: 0.8667

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1. Problem Statement

1.1 Business Problem

The objective of this project is to develop a machine learning model that can predict medical insurance charges for an individual based on demographic and lifestyle-related information. Insurance charges vary according to factors such as age, body mass index, number of children, gender and smoking status. A regression model can learn these relationships from historical data and estimate the expected insurance charge for a new individual.

1.2 Machine Learning Problem

This project is a supervised learning regression problem because the target variable, charges, is a continuous numerical value. Several regression algorithms were developed and compared using the R^2 score. The model with the highest test-set R^2 score was selected as the final model.

Item	Description
Problem Type	Supervised Learning – Regression
Objective	Predict medical insurance charges
Target Variable	charges
Evaluation Metric	R^2 Score
Models Evaluated	Multiple Linear Regression, SVR, Decision Tree, Random Forest

2. Dataset Information

The project uses an insurance dataset containing individual-level information about age, sex, BMI, number of children, smoking status and medical insurance charges. The dataset contains 1,338 records and 6 columns.

Column	Data Type / Nature	Role
age	Numerical	Input feature
sex	Categorical	Input feature
bmi	Numerical	Input feature
children	Numerical	Input feature
smoker	Categorical	Input feature
charges	Numerical / Continuous	Target

2.1 Dataset Snapshot

	age	sex	bmi	children	smoker	charges
	19	female	27.9	0	yes	16884.92
	18	male	33.77	1	no	1725.552
	28	male	33	3	no	4449.462
	33	male	22.705	0	no	21984.47
	32	male	28.88	0	no	3866.855
	31	female	25.74	0	no	3756.622
	46	female	33.44	1	no	8240.59
	37	female	27.74	3	no	7281.506
0	37	male	29.83	2	no	6406.411
1	60	female	25.84	0	no	28923.14
2	25	male	26.22	0	no	2721.321
3	62	female	26.29	0	yes	27808.73
4	23	male	34.4	0	no	1826.843
5	56	female	39.82	0	no	11090.72
6	27	male	42.13	0	yes	39611.76
7	19	male	24.6	1	no	1837.237
8	52	female	30.78	1	no	10797.34

Figure 1: Sample view of the insurance dataset.

3. Data Pre-processing

3.1 Converting Categorical Variables

The original dataset contains categorical variables such as sex and smoker. These variables cannot be directly supplied to the regression algorithms in their original text form. Therefore, one-hot encoding was performed using pandas `get_dummies()`. The `drop_first=True` option was used to remove the redundant first category.

```
dataset = pd.get_dummies(dataset, drop_first=True)
dataset = dataset.astype(int)
```

After preprocessing, the categorical variables were represented numerically as `sex_male` and `smoker_yes`. The resulting model inputs were age, bmi, children, `sex_male` and `smoker_yes`, while charges remained the dependent variable.

3.2 Feature and Target Selection

```
independent = dataset[['age', 'bmi', 'children', 'sex_male', 'smoker_yes']]
dependent = dataset[['charges']]
```

4. Train-Test Split

The dataset was divided into training and testing subsets using a 70:30 split. The training set was used to build the models and the test set was used to evaluate their predictive performance. A `random_state` of 0 was used to make the split reproducible.

```
X_train, X_test, Y_train, Y_test = train_test_split(
    independent, dependent, test_size=0.30, random_state=0
)
```

Setting	Value
Training data	70%
Testing data	30%
Random state	0

5. Multiple Linear Regression

Multiple Linear Regression was used as a baseline model. It estimates the relationship between the insurance charges and multiple independent variables simultaneously.

Result

Test R^2 Score = 0.7893

The model achieved an R^2 score of 0.7893 on the test data. This provided a useful baseline for comparing the performance of the other regression algorithms.

6. Support Vector Regression (SVM)

Support Vector Regression (SVR) was evaluated using four kernels: linear, rbf, poly and sigmoid. The C parameter was varied across six values: 1, 10, 100, 1000, 2000 and 3000. A total of 24 parameter combinations were evaluated.

S.No	C	Linear	RBF	Poly	Sigmoid
1	C=1	-0.1115	-0.0884	-0.0646	-0.0899
2	C=10	-0.0017	-0.0818	-0.0931	-0.0909
3	C=100	0.5432	-0.1245	-0.0992	-0.1185
4	C=1000	0.6339	-0.1176	-0.0547	-1.7112
5	C=2000	0.6899	-0.1078	-0.0016	-5.8191
6	C=3000	0.7590	-0.0962	0.0494	-12.5445

Best SVM Configuration

The highest SVM R^2 score was 0.7590, obtained using the linear kernel with C = 3000.

7. Decision Tree Regression

Decision Tree Regression was evaluated using three criteria, two splitter options and three max_features settings. This produced 18 combinations.

S.No	Criterion	Splitter	Max_features	R^2
1	squared_error	best	None	0.6909
2	squared_error	best	sqrt	0.7839
3	squared_error	best	log2	0.7839
4	squared_error	random	None	0.7263
5	squared_error	random	sqrt	0.6793
6	squared_error	random	log2	0.6793
7	absolute_error	best	None	0.7310
8	absolute_error	best	sqrt	0.7072
9	absolute_error	best	log2	0.7072
10	absolute_error	random	None	0.7134
11	absolute_error	random	sqrt	0.7065
12	absolute_error	random	log2	0.7065
13	poisson	best	None	0.6946
14	poisson	best	sqrt	0.7396
15	poisson	best	log2	0.7396
16	poisson	random	None	0.7161
17	poisson	random	sqrt	0.6834
18	poisson	random	log2	0.6833

Best Decision Tree Configuration

The highest Decision Tree R^2 score was 0.7839. It was achieved by squared_error + best + sqrt and also by squared_error + best + log2.

8. Random Forest Regression

Random Forest Regression was evaluated by varying criterion, max_features and n_estimators. The three criteria were squared_error, absolute_error and poisson; max_features was tested using 1.0, sqrt and log2; and n_estimators was tested using 10, 50 and 100. This resulted in 27 combinations.

S.No	Criterion	Max_features	n_estimators	R ²
1	squared_error	1.0	10	0.8254
2	squared_error	1.0	50	0.8519
3	squared_error	1.0	100	0.8525
4	squared_error	sqrt	10	0.8414
5	squared_error	sqrt	50	0.8604
6	squared_error	sqrt	100	0.8662
7	squared_error	log2	10	0.8414
8	squared_error	log2	50	0.8604
9	squared_error	log2	100	0.8662
10	absolute_error	1.0	10	0.8412
11	absolute_error	1.0	50	0.8577
12	absolute_error	1.0	100	0.8566
13	absolute_error	sqrt	10	0.8442
14	absolute_error	sqrt	50	0.8621
15	absolute_error	sqrt	100	0.8667
16	absolute_error	log2	10	0.8442
17	absolute_error	log2	50	0.8621
18	absolute_error	log2	100	0.8667
19	poisson	1.0	10	0.8280
20	poisson	1.0	50	0.8496
21	poisson	1.0	100	0.8505
22	poisson	sqrt	10	0.8366
23	poisson	sqrt	50	0.8548
24	poisson	sqrt	100	0.8591
25	poisson	log2	10	0.8366
26	poisson	log2	50	0.8548
27	poisson	log2	100	0.8591

Best Random Forest Configuration

The highest observed R² score was 0.8667. Two configurations tied at this value: absolute_error + sqrt + 100 estimators and absolute_error + log2 + 100 estimators. For the final model, the sqrt configuration is selected because it is the first tied configuration in the documented experiment and provides the same predictive score.

```
RandomForestRegressor(  
    criterion='absolute_error',  
    max_features='sqrt',  
    n_estimators=100,  
    random_state=0  
)
```

9. Model Comparison

Model	Best Configuration	Best R ² Score
Multiple Linear Regression	Baseline	0.7893
Support Vector Regression	linear, C=3000	0.7590
Decision Tree Regression	squared_error, best, sqrt	0.7839
Random Forest Regression	absolute_error, sqrt, 100	0.8667

The comparison shows that Random Forest Regression achieved the **highest R² score of 0.8667**. Its performance was higher than Multiple Linear Regression (0.7893), Decision Tree Regression (0.7839), and Support Vector Regression (0.7590).

10. Final Model Selection and Justification

The final model selected for the insurance charges prediction problem is the Random Forest Regressor with **criterion='absolute_error', max_features='sqrt', and n_estimators=100**. This configuration achieved the **highest test R² score of 0.8667** among all the evaluated models and hyperparameter combinations.

Random Forest is an ensemble learning method that combines multiple decision trees and aggregates their predictions. This allows the model to capture non-linear relationships between the input variables and insurance charges and generally provides more robust predictions than relying on a single decision tree.

The model was selected based on experimental evidence rather than only theoretical preference. The highest observed R² score was used as the main selection criterion, and the complete research values were documented in the preceding tables.

Final Model: Random Forest Regressor | Test R² = 0.8667

11. Conclusion

This project developed a machine learning solution for predicting medical insurance charges. The dataset contained 1,338 records and six columns. Categorical variables were converted into numerical dummy variables using one-hot encoding, after which the features and target variable were separated and the data was divided into training and testing sets using a 70:30 ratio.

Four regression approaches were evaluated: Multiple Linear Regression, Support Vector Regression, Decision Tree Regression and Random Forest Regression. Different hyperparameter combinations were systematically tested for the latter three models, and the R² scores were documented.

The Random Forest Regressor achieved the best test R² score of 0.8667 using criterion='absolute_error', max_features='sqrt' and n_estimators=100. Therefore, this configuration was selected as the final model for predicting insurance charges.

Result Summary

Item	Final Result
Best Model	Random Forest Regression
Criterion	absolute_error
Max Features	sqrt
Number of Estimators	100
Best Test R ² Score	0.8667